QP CODE: 25900317

THIRD SEMESTER MGU-BCA (HONOURS) REGULAR EXAMINATION
OCTOBER 2025
Core Course (CC)-MG3CCRBCA203-DESIGN AND ANALYSIS OF ALGORITHM
(2024 ADMISSION ONWARDS)

SCEHEME FOR VALUATION

Time: 1.50 hours **Maximum Marks: 50**

Part-A
Very Short Answer Questions. Answer All Questions
Each question carries 2 marks

1. **Given the code, give the space complexity (in terms of auxiliary space):**
   function f(n)
   if n == 0:return
   temp = array of size n
   return f(n-1) + temp **[An] / [CO1]**
   Answer:
   Each recursive call creates an array of size n, and there are n recursive calls.
   Total space used = $1 + 2 + 3 + \ldots + n = n(n + 1)/2 = O(n^2)$.
   Calculation of auxiliary space of above code – 2 marks
2. **Define Recursion.** **[K] / [CO1]**
   Answer: Recursion is a programming technique in which a function calls itself directly or indirectly to solve a smaller instance of the same problem.
3. **What is the advantage of Quick Sort over Merge Sort?** **[U] / [CO2]**
   Answer: Merge Sort requires extra memory to create temporary subarrays, so its space complexity is O(n). Quick Sort is an in-place algorithm and needs only O(log n) space for recursive calls. Although both have the same average time complexity O(n log n), Quick Sort is usually faster in practice because it uses memory efficiently and has better cache performance.
4. **Discuss the worst-case complexity of binary search.** **[U] / [CO2]**
   Answer: In Binary Search, the search space is divided by 2 after each comparison. The worst-case occurs when the element is not present or found at the last step. Hence, the worst-case time complexity - O(log n).
5. **Why does Strassen's algorithm use addition and subtraction operations along with multiplication?** **[U] / [CO2]**
   Answer: Matrix multiplication is more expensive than addition or subtraction. Strassen's algorithm reduces the number of multiplications from 8 to 7 by reformulating the matrix equations and using 18 addition and subtraction operations. This makes the overall computation faster and more efficient.
6. **Explain how the Divide and Conquer technique is applied to find the maximum and minimum values in a list of elements.** [U] / [CO2]

Answer: The MaxMin algorithm uses the Divide and Conquer approach by dividing the list into two halves. For single or two elements, the maximum and minimum are found directly. For larger sets, the midpoint is determined, and the problem is solved recursively for both halves. Finally, the two maxima and two minima are compared to obtain the overall maximum and minimum values.

7. **Define biconnected components**. [**K] / [CO3]**

Answer: A biconnected component is a maximal subgraph of a graph that contains no articulation points**.** Or it is a subgraph that remains connected even after the removal of any single vertex**.**

8. **Define principle of optimality [K] / [CO3]**

Answer: The principle of optimality states that an optimal sequence of decisions has the property that whatever the initial state and decision are, the remaining decisions must constitute an optimal decision sequence with regard to the state resulting from the first decision.

9. **List any two characteristics of BFS**. **[K] / [CO3]**

Answer: In Breadth First Search (BFS), we start from a vertex, mark it as visited, and then explore all its adjacent unvisited vertices. The list of unexplored vertices is maintained using a queue, ensuring vertices are explored level by level.

10. **Define Backtracking. [K] / [CO3]**

Answer: Backtracking is a problem-solving technique used to find solutions by building them step by step. It explores all possible combinations of values (x_1, x_2, …, x_n) that satisfy given explicit and implicit constraints, and backtracks whenever a partial solution violates any constraint**.**

(10 X 2 = 20)

Part-B

Short Answer Questions. Answer Any 6 Questions.

Each question carries 5 marks

11. **List out the advantages of using algorithm. [U] / [CO1]**

Answer:

i. Efficiency: Solves problems using minimum time and resources.
ii. Definiteness: Each instruction is clear and unambiguous.
iii. Finiteness: Always terminates after a finite number of steps.
iv. Correctness: Produces the desired output for valid inputs.
v. Reusability: Can be applied to similar problems without redesign.
vi. Optimization: Helps to choose the most efficient solution.
vii Independence: Not dependent on any specific programming language.

Explain any five points - one mark each

12. **Explain the Divide conquer general method? [U] / [CO2]**

Answer:

Explanation of divide and conquer method with control abstraction -3 marks
Example -2 marks

13. **List the major steps in designing an algorithm using the greedy method. [K] / [CO2]**

Answer:

- Select an input according to a specific optimization measure (objective function).

- Check whether including the selected input keeps the solution feasible.
- If feasible, include the element in the partially constructed optimal solution; otherwise, reject it. Continue the process for all inputs according to the selection order
- When all inputs are considered, the constructed set gives the greedy (approximate or optimal) solution. *– 4 marks + Example – 1 mark*

Or

Explanation with control abstraction of Greedy method (4 Marks) with an example (1 Mark)

14. Write the step-by-step procedure of Dijkstra's Algorithm. [U] / [CO2]

Answer:

Explanation of Dijkstra's Algorithm with an example – (3+2 Marks)

15. Explain the Sum of Subsets problem with an algorithm. Solve it using the data set S = {3, 5, 6, 7} with d = 15. [An] / [CO3]

Answer: Apply procedure of sum of sets the final solution is {3, 5, 7}

Explanation of algorithm of sum of subset problem – 3 marks.

Solution for given dataset - 2 Marks

[5 x 3 = 15]

Part C

Essay Questions. Answer any one question.

Each question carries 15 marks

16. Discuss in detail the working mechanism, data structures used, and efficiency of Prim's algorithm. Illustrate with a suitable example. [An] / [CO2]

Answer:

Definition & Purpose of Prim's Algorithm- 2 Marks

Algorithm/ Explanation - 6 Marks,

Data Structures Used - 2 Marks

Efficiency / Complexity - 2 Marks,

MST Construction with example -3 Marks

17. Apply Hamiltonian cycle algorithm to find cycles in a 6-vertex complete graph. [A] / [CO3]

Answer:

Definition of Hamiltonian cycle- 2 Marks

Algorithm and Explanation -8 Marks,

Example of 6 vertex graph with solution – 5 Marks

[15 x 1 = 15]

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